ENDG 319

Probability, Statistics and Machine Learning

Lecture 10: More Discrete Distributions

Poisson • Hypergeometric • Geometric
Negative binomial • Multinomial

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Learning outcomes for Lecture 10

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DESCRIBE THE EXPERIMENT BEFORE SELECTING THE DISTRIBUTION

1	**Recognize**	Distinguish fixed-opportunity counts, finite samples, and stopping rules.
2	**Parameterize**	Identify λ, N, R, n, r, and p from their meanings—not just their values.
3	**Calculate**	Use each PMF and determine its allowed count values.
4	**Interpret**	Explain means, variances, and the assumptions behind the model.

> “Success” is a label; the sampling and stopping rules determine the distribution.

From binomial counts to a Poisson approximation

MANY TRIALS • SMALL SUCCESS PROBABILITY

Binomial starting point

$$X \sim \text{Bin}(n, p)$$

n is large; p is small.

Match the mean count

$$\lambda = np$$

Use a Poisson model with this λ.

$$\binom{n}{x} p^x (1-p)^{n-x} \approx e^{-\lambda} \frac{\lambda^x}{x!}$$

This is an approximation—not an identity. The Poisson parameter is the expected count.

Poisson experiments count events in an opportunity

THE OPPORTUNITY MAY BE TIME, LENGTH, AREA, OR VOLUME

Time interval

X = number of customers served in 5 minutes



Length interval

X = number of defects in a 5 cm piece of wire



Counting in an interval does not guarantee Poisson; it remains a model assumption.

The Poisson probability mass function

λ IS A MEAN COUNT; x IS A POSSIBLE OBSERVED COUNT

$$X \sim \text{Poisson}(\lambda), \quad \lambda > 0$$

$$p(x) = P(X = x) = e^{-\lambda}\frac{\lambda^x}{x!}$$

Allowed values $x = 0, 1, 2, \ldots$

Outside these values $p(x) = 0$

λ is a mean, not a probability. x must be a nonnegative integer.

Poisson mean, variance, and opportunity size

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MATCH THE MEAN TO THE INTERVAL IN THE QUESTION

MEAN	VARIANCE	STANDARD DEVIATION
$\mu_X = \lambda$	$\sigma_X^2 = \lambda$	$\sigma_X = \sqrt{\lambda}$

From an intensity to a mean count

$$\lambda = (\text{events per unit})(\text{number of units})$$

Mean and variance have the same numerical value under the Poisson model; SD is $\sqrt{\lambda}$.

Example 3: particles in a withdrawn volume

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ORIGINAL QUESTION • DEFINE THE RANDOM VARIABLE FIRST

Particles are suspended in a liquid at a concentration of 6 particles per mL. A large volume is thoroughly agitated, and then 3 mL are withdrawn.

Concentration	Withdrawn volume	Requested count
6 particles/mL	**3 mL**	**15 particles**

What is the probability that exactly 15 particles are withdrawn?

X counts particles in the 3 mL sample; apply the assumed Poisson model.

Example 3: scale the mean, then use the PMF

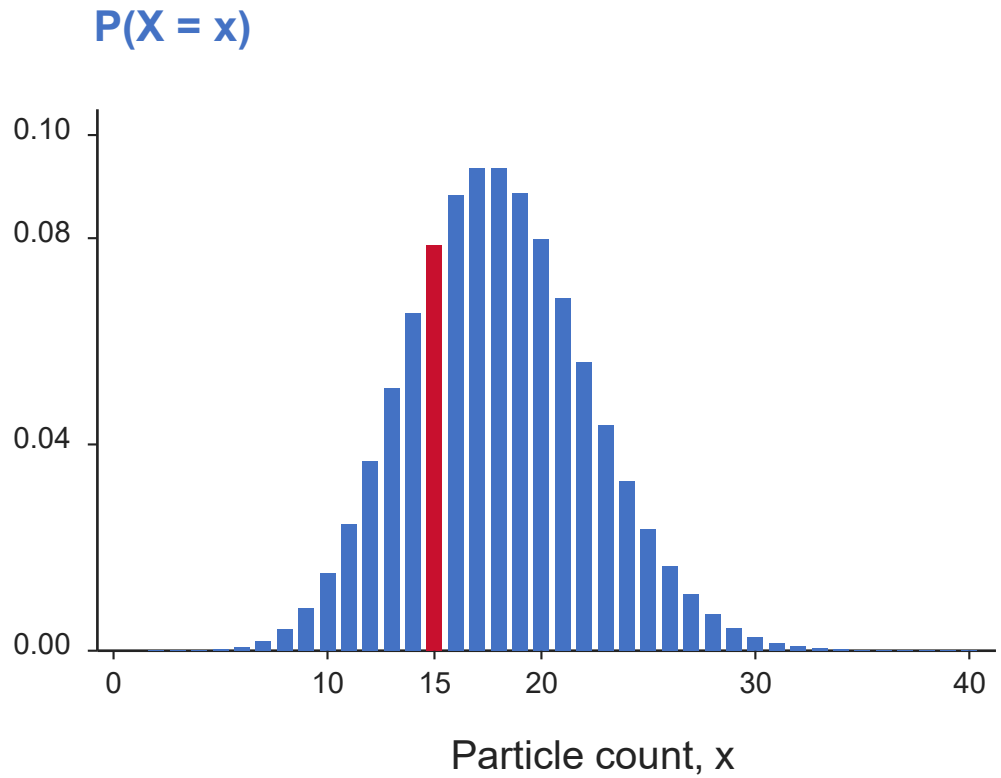
WORKED SOLUTION DERIVED FROM THE SUPPLIED INPUTS

$$\lambda = 6 \times 3 = 18$$

$$P(X = 15) = e^{-18}\frac{18^{15}}{15!}$$

≈ 0.0786 = 7.86%

The mean is 18 particles;
the event asks for exactly 15.



Using λ = 6 would describe a 1 mL opportunity, not the requested 3 mL sample.

Three checks before a Poisson calculation

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OPPORTUNITY → PARAMETER → EVENT

1 Specify the opportunity

Is the count per minute, per 5 cm, or in the withdrawn volume?

2 Compute the mean for it

λ describes that complete opportunity, not necessarily one unit.

3 Translate the event

“Exactly x” uses one mass; do not replace x with the mean.

> Before calculating, write what X counts and what λ means in one complete sentence.

Hypergeometric: a finite sample without replacement

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A FIXED POPULATION • TWO TYPES • A SIMPLE RANDOM SAMPLE

Population

N items

R successes

N − R failures

Sample

n distinct items

X = number of successes among the n sampled items

H(N, R, n): N = population size; R = population successes; n = sample size.

Random selection does not make the draws independent

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UPDATE BOTH THE SUCCESS COUNT AND THE POPULATION SIZE

$$P(\text{success on first draw}) = \frac{R}{N}$$

After a success

$$P(S_2 \mid S_1) = \frac{R-1}{N-1}$$

One success and one item have been removed.

After a failure

$$P(S_2 \mid S_1^c) = \frac{R}{N-1}$$

The success count stays R; one item is removed.

The next-draw probability depends on what was removed; the draws are not independent.

Build the hypergeometric PMF by counting subsets

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CHOOSE SUCCESSES AND FAILURES, THEN DIVIDE BY ALL SAMPLES

Success choices	Failure choices	All n-item samples
$\binom{R}{x}$	$\binom{N-R}{n-x}$	$\binom{N}{n}$

$$p(x) = P(X = x) = \frac{\binom{R}{x}\binom{N-R}{n-x}}{\binom{N}{n}}$$

Use unordered subsets throughout; x successes and n − x failures make n items.

Hypergeometric support: which counts are feasible?

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THE POPULATION LIMITS BOTH SUCCESSES AND FAILURES

Requirement	Consequence
A count cannot be negative.	$x \geq 0$
No more than n sampled items can be successes.	$x \leq n$
Only R successes are available.	$x \leq R$
The n − x failures must come from N − R items.	$x \geq n - (N - R)$

$$\max(0, R + n - N) \leq x \leq \min(n, R)$$

x must be an integer in this range. The PMF is zero at every other value.

Hypergeometric mean and variance

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SAMPLING WITHOUT REPLACEMENT REDUCES THE COUNT VARIANCE

Mean

$$\mu_X = n\frac{R}{N}$$

Variance

$$\sigma_X^2 = n\frac{R}{N}\left(1-\frac{R}{N}\right)\frac{N-n}{N-1}$$

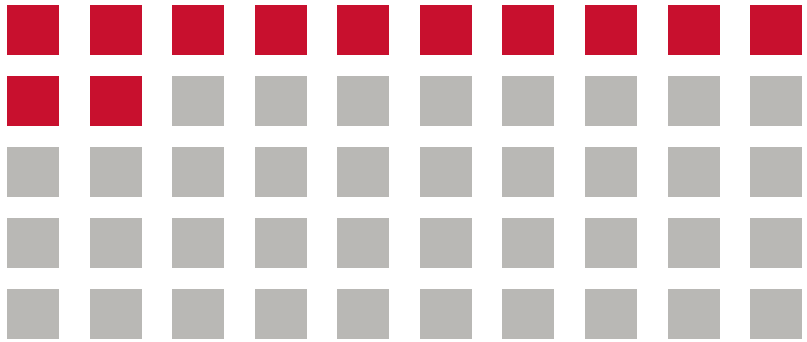
Finite-population factor	$\frac{N-n}{N-1}$	Use N > 1.

The mean matches Bin(n, R/N); the variance includes a finite-population factor.

Example 4: inspecting buildings for code violations

ORIGINAL QUESTION • COUNT "VIOLATION" AS A SUCCESS

Of 50 buildings in an industrial park, 12 have electrical code violations. Select 10 buildings at random for inspection.



12 with violations + 38 without

Find:

P(X = 3)
The mean of X
The variance of X

X counts violations among 10 inspected buildings: X ~ H(50, 12, 10).

Example 4: count samples with exactly three violations

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WORKED SOLUTION DERIVED FROM THE SOURCE PMF

Select 3 of the 12 violation buildings and 7 of the 38 others.

$$P(X=3)=\frac{\binom{12}{3}\binom{38}{7}}{\binom{50}{10}}$$

≈ 0.2703 = 27.03%

Allowed values in this example: x = 0, 1, …, 10.

The denominator counts all 10-building subsets—not individual buildings.

Example 4: mean and variance of the violation count

SAME PARAMETERS; USE THE FINITE-POPULATION FACTOR

Mean

$$\mu_X = 10\left(\frac{12}{50}\right) = 2.4$$

Variance

$$\sigma_X^2 = 10\left(\frac{12}{50}\right)\left(\frac{38}{50}\right)\left(\frac{40}{49}\right)$$

$$= \frac{1824}{1225} \approx 1.4890$$

Without the finite-population factor: 1.824—not the required variance.

A mean of 2.4 describes repeated samples; each actual count is an integer.

Geometric: stop at the first success

INDEPENDENT BERNOULLI TRIALS WITH THE SAME SUCCESS PROBABILITY p

X is the number of trials up to and including the first success.

F	F	...	F	S
Trial 1	Trial 2		Trial x − 1	Trial x

$$X \sim \text{Geom}(p), \qquad x = 1, 2, 3, \ldots$$

The number of trials is random. The final successful trial is included in X.

Geometric PMF, mean, and variance

x − 1 FAILURES FOLLOWED BY ONE SUCCESS

$$P(X=x)=(1-p)^{x-1}p, \qquad x=1,2,\ldots$$

p(x) = 0 otherwise; 0 < p ≤ 1.

MEAN

$$\mu_X=\frac{1}{p}$$

VARIANCE

$$\sigma_X^2=\frac{1-p}{p^2}$$

Do not use x failures: trial x is the first success, so only x − 1 trials are failures.

Example 5: audit until an invoice qualifies

ORIGINAL QUESTION • THE SUCCESS IS “QUALIFIES FOR A DISCOUNT”

A large industrial firm allows a discount on any invoice paid within 30 days. Twenty percent of invoices receive the discount.

Invoices are audited one by one. Let X be the number audited up to and including the first invoice that qualifies for a discount.

What is the distribution of X?

The geometric model assumes independent invoice outcomes with constant p = 0.20.

Example 5: identify the model and its support

WORKED SOLUTION UNDER THE STATED INDEPENDENCE ASSUMPTION

$$X \sim \text{Geom}(0.20)$$

$$P(X = x) = 0.20(0.80)^{x-1}, \qquad x = 1, 2, \ldots$$

Zero probability at all other values.

Derived moment check $\mu_X = 5, \qquad \sigma_X^2 = 20$

A mean of 5 audits does not guarantee that the first qualifying invoice is number 5.

Negative binomial: stop at the rth success

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EXTEND THE GEOMETRIC STOPPING RULE FROM ONE SUCCESS TO r

Conduct independent Bernoulli trials with the same success probability p.
Fix a positive integer r.

X counts all trials up to and including success r.

Wait for success 1	Wait for success 2	…	Wait for success r

$$X \sim \mathrm{NB}(r,p), \qquad x = r, r+1, \ldots$$

r is the fixed target number of successes; X is the random total number of trials.

Why the negative-binomial coefficient uses x − 1

THE LAST TRIAL MUST BE THE rth SUCCESS

First x − 1 trials

Exactly r − 1 successes and x − r failures

Trial x

SUCCESS

$$P(X=x)=\binom{x-1}{r-1}p^r(1-p)^{x-r}$$

x = r, r + 1, … ; p(x) = 0 otherwise; $0 < p \leq 1$.

Using C(x, r) would also count sequences in which the rth success occurs before trial x.

Negative-binomial moments from geometric waiting times

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ADD r INDEPENDENT WAITS FOR THE NEXT SUCCESS

$$X = Y_1 + Y_2 + \cdots + Y_r$$

$Y_1, \ldots, Y_r$ are independent, each with Geom(p) distribution.

MEAN

$$\mu_X = r\left(\frac{1}{p}\right) = \frac{r}{p}$$

VARIANCE

$$\sigma_X^2 = r\frac{1-p}{p^2}$$

When r = 1, both the PMF and the moments reduce to those of Geom(p).

Fixed trials versus a stopping rule

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BINOMIAL, GEOMETRIC, AND NEGATIVE BINOMIAL SHARE THE TRIAL ASSUMPTIONS

Model	What is fixed?	What does X count?	Possible X
Binomial	n trials	Successes in n trials	0, …, n
Geometric	First success	Trials through that success	1, 2, …
Negative binomial	rth success	Trials through that success	r, r + 1, …

All three use independent Bernoulli trials with a common success probability p.

> Do not choose a distribution from p alone. First identify what ends the experiment.

Multinomial trials: one of k categories on each trial

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GENERALIZE A BERNOULLI TRIAL FROM TWO OUTCOMES TO k ≥ 2

Each trial produces exactly one category. Across n independent trials, the category probabilities remain the same.

Category 1	Category 2	…	Category k
p_1	p_2		p_k

$$p_1 + \cdots + p_k = 1, \qquad p_i \geq 0$$

The categories are mutually exclusive and exhaustive: exactly one applies per trial.

The multinomial joint probability mass function

A COUNT VECTOR THAT MUST ADD TO THE FIXED TOTAL n

$$(X_1, \ldots, X_k) \sim \mathrm{MN}(n, p_1, \ldots, p_k)$$

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} \prod_{i=1}^{k} p_i^{x_i}$$

$$x_i \in \{0, 1, \ldots, n\}, \qquad \sum_{i=1}^{k} x_i = n$$

The joint PMF is zero outside these count constraints.

Corrected notation: k counts categories; each category count ranges from 0 to n.

Binomial marginals do not make the counts independent

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INDEPENDENT TRIALS ARE NOT INDEPENDENT CATEGORY COUNTS

$$X_i \sim \text{Bin}(n, p_i)$$

For one category i, each trial is either “category i” or “not category i.”

$$X_1 + \cdots + X_k = n$$

If k − 1 category counts are known, the remaining count is determined.

Binomial marginals do not justify multiplying them to obtain the joint probability.

Example 6: classify four assembly-line items

ORIGINAL QUESTION • INDEPENDENT ITEMS, THREE CLASSIFICATIONS

Each inspected item is classified as conforming (acceptable), downgraded, or rejected. Four items are chosen independently and at random.

Category	Conforming	Downgraded	Rejected
Probability	0.70	0.20	0.10
Count among 4	X_1	X_2	X_3

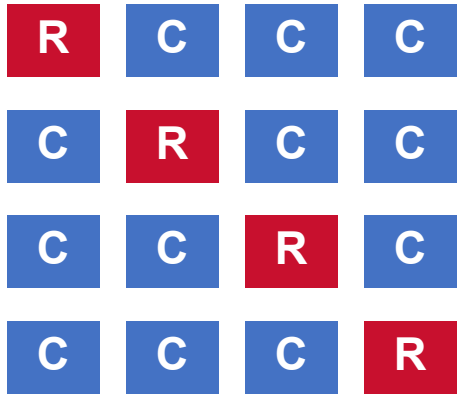
1. What is the joint distribution of X_1, X_2, X_3?
2. What is the probability of 3 conforming and 1 rejected?

Use $(x_1, x_2, x_3) = (3, 0, 1)$: the event includes no downgraded items.

Example 6: four arrangements contribute to the event

WORKED SOLUTION • ONE REJECTION CAN OCCUR IN FOUR POSITIONS

$$(X_1, X_2, X_3) \sim \text{MN}(4, 0.70, 0.20, 0.10)$$



C = conforming; R = rejected

$$p(3, 0, 1) = \frac{4!}{3!\,0!\,1!}(0.70)^3(0.20)^0(0.10)$$

$$= 4(0.70)^3(0.10) = 0.1372$$

13.72%

Four sequences share these totals. Omitting the factor 4 counts only one sequence.

Choose the distribution from the experiment

FIRST ASK WHAT IS FIXED AND WHAT IS BEING COUNTED

Experiment	Random quantity	Model
n independent binary trials, same p	Success count	Binomial
Count in a fixed time or space opportunity	Event count	Poisson model
n items from a finite population, no replacement	Sample success count	Hypergeometric
Independent trials until the first success	Total trial count	Geometric
Independent trials until success r	Total trial count	Negative binomial
n independent trials with k categories	Vector of category counts	Multinomial

Choose the model from the experiment, assumptions, and definition of X.

Keep parameter meanings beside the moment formulas

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MEANS AND VARIANCES SUMMARIZE DIFFERENT COUNTING EXPERIMENTS

Model	Mean	Variance
Poisson(λ)	λ	λ
H(N, R, n)	nR/N	$n\frac{R}{N}(1-\frac{R}{N})\frac{N-n}{N-1}$
Geom(p)	$1/p$	$(1-p)/p^2$
NB(r, p)	r/p	$r(1-p)/p^2$
MN: each X_i	np_i	$np_i(1-p_i)$

The multinomial row uses binomial marginals; the category counts are not independent.

Common mistakes: identify the quantity before calculating

A CORRECT FORMULA CAN STILL BE APPLIED TO THE WRONG EXPERIMENT

Poisson	Confusing a per-unit rate with λ for the complete opportunity.
Hypergeometric	Treating no-replacement draws as independent or ignoring support.
Geometric	Counting failures only, although X includes the final successful trial.
Negative binomial	Using C(x, r) instead of fixing success r at trial x.
Multinomial	Ignoring the zero-count category or assuming category counts are independent.

Check the experiment, parameters, support, and stopping rule before calculating.

Concept check: identify and repair each incorrect claim

DISCUSS BEFORE REVEALING THE NEXT SLIDE

A In Example 3, use $\lambda = 6$ because the concentration is 6 particles/mL.

B The building selections are independent because they are made at random.

C For NB(r, p), trial x may fail even when X = x.

D In the multinomial example, X_1, X_2, and X_3 are independent because the four trials are independent.

For each statement, name the violated definition or assumption—not only “false.”

Concept check: the corrected reasoning

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REPAIR THE MODEL BEFORE REPAIRING THE CALCULATION

A **$\lambda = 6 \times 3 = 18$.**

Use the expected count in the requested 3 mL volume.

B **No replacement creates dependence.**

The remaining success fraction depends on what was selected.

C **Trial x must be a success.**

The first $x - 1$ trials contain exactly $r - 1$ successes.

D **Trial independence is not count independence.**

$X_1 + X_2 + X_3 = 4$ constrains the three totals.

A model is useful only when the random variable and its assumptions match the question.

Lecture 10: key takeaways and questions

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MATCH THE RANDOM VARIABLE TO THE WAY THE DATA ARE GENERATED

Poisson	Count events in a specified opportunity; match λ to that opportunity.
Hypergeometric	Count successes in a finite sample drawn without replacement.
Geometric / NB	Count trials through the first / rth success; include the final success.
Multinomial	Count several categories across fixed n independent trials; totals add to n.

Questions?

> Define X → check assumptions → identify parameters and support → calculate.